Exact inputs behind the reported structural comparisons

Modeling choice	Step 1: activity panel	Step 2: activity panel	Step 1: mutant scan
	48 homologs + SfaB reference	Same 49 enzymes	AS9 / AS17 / AS23; matched WT
Chemical state	ATP (CCD) + adipate; pre-activation complex	Adipyl-AMP intermediate + HMDA; neutral attacking amine	ATP (CCD) + adipate; pre-activation complex
Formal charges	Adipate: input -2; saved processed graph 0^{*}	Adipyl-AMP -2; HMDA $+1$ with one NH_2 and one NH_3^+	Adipate -2 input; cached CCD ATP $0^{\dagger}$
Metal / protein	1 Mg^{2+}; one protein chain	1 Mg^{2+}; one protein chain	2 Mg^{2+}; one protein chain
Spatial restraints	None; adipate requested as affinity binder*	Forced HMDA–pocket restraint: maximum distance 10 Å	None; no affinity property
MSA	Cached sequence-specific MSA	Same protein MSA as step 1	Separate MSA for each mutant; query sequence verified
Sampling	Boltz-2; 50 diffusion samples; no explicit seed in launcher	Boltz-2; 50 diffusion samples; no explicit seed in launcher	50 samples; seed 20261019; 3 recycles; 200 sampling steps
Reported quantity	Proximity-weighted O–P_α–O3A angle; all 50 poses	Neutral amine-N to activated acyl-C distance	Change in weighted angle versus fresh two-Mg WT

Reaction products determine the activation coordinate

AMP-forming: acyl-adenylate (e.g., DdaG)
Carboxyl O attacks P_α; leaving bond P_α–O3A
ATP → AMP + PP_i | $\theta = \angle(O, P_\alpha, O3A)$

ADP-forming: acyl-phosphate (e.g., PuuA)
Carboxyl O attacks P_γ; leaving bond P_γ–O3B
ATP → ADP + P_i | $\theta = \angle(O, P_\gamma, O3B)$

Discovery ranking (slide 17): recorded inputs use 1 or 2 Mg ions by candidate; score the assigned P_α or P_γ coordinate. Mg stoichiometry is an input assumption, independent of coordinate assignment.

Sensitivity remains to be tested: compare within matched setups; the one-Mg panel does not validate transfer to the two-Mg mutant scan. Pose spread measures prediction variability.

*All 49 step-1 processed ligand graphs are neutral despite charged input SMILES; the installed parser standardizes affinity ligands before construction. †The mutant campaign audit records CCD ATP charge 0; do not label it ATP^{4-}. Historical CCD identity is not pinned for every panel run.

Sources: saved panel YAMLs and ligand graphs; mutant campaign chemistry audit, inputs and launcher; discovery coordinate audit. Full identifiers, hashes and speaker notes accompany this slide.